

Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #42: Internal Microservice Catalog & Health Portal

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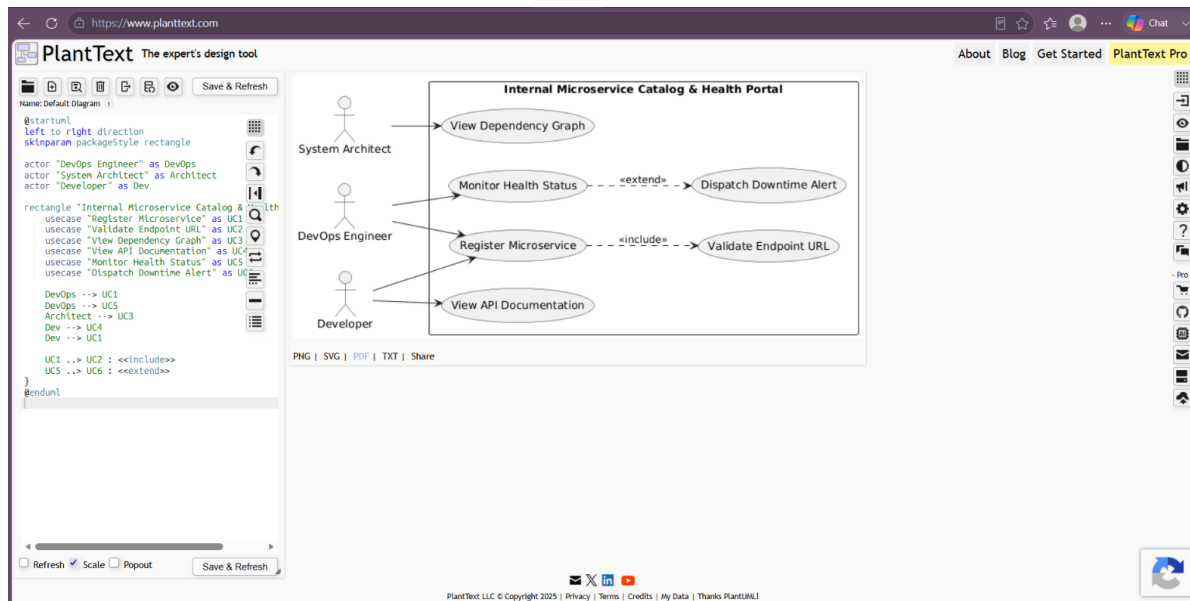
SRN: PES1UG24AM440

Section: H

1. Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ping microservice health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.	High	Pass: Downtime recorded and alert dispatched on 3 consecutive failed health probes; Fail: Unreachable service marked healthy.	Ensures accurate tracking of service uptime.
FR-002	Functional	The system shall allow a System Architect to view a visual dependency graph of all registered microservices.	High	A graph is successfully rendered showing nodes and their active connections when the architect accesses the dashboard.	Fulfills the primary catalog mapping objective.
FR-003	Functional	The system shall aggregate and display API documentation submitted for each microservice.	Medium	API documentation text is visible and searchable on the individual service details page.	Centralizes enterprise developer resources.
FR-004	Functional	The system shall allow a DevOps Engineer to manually trigger an immediate health check on a selected microservice.	Medium	The system immediately pings the endpoint and updates the status UI without waiting for the 30-second interval.	Allows for immediate validation after a deployment or fix.
FR-005	Functional	The system shall allow users to register a new microservice by inputting its name, health endpoint URL, and API documentation link.	High	A newly submitted service appears in the catalog database and is immediately added to the health ping schedule.	Required to populate the catalog dynamically.
NFR-001	Nonfunctional	The service catalog dependency graph viewer must render interactive node architectures containing up to 200 services smoothly.	High	Benchmarking tests confirm target latency and security standards under simulated peak load.	Prevents portal UI freezing during complex architecture reviews.
NFR-002	Nonfunctional	The system shall dispatch downtime alerts to the specified DevOps Engineer within 5 seconds of the third consecutive failed health probe.	High	System logs show the timestamp delta between the third failure and alert dispatch is < 5 seconds.	Ensures rapid incident response for critical outages.

2. UML Use-case diagram:



3. Use-Case Flow Document:

Use Case: Register Microservice

Actor: Developer, DevOps Engineer

Main Success Scenario:

1. The user navigates to the portal and selects Register New Service.
2. The system displays a registration form requesting the service name, description, health endpoint URL, and API documentation link.
3. The user inputs all required fields and clicks Submit.

4. The system validates the format of the provided URLs («include» relationship).
5. The system performs an initial ping to the provided health endpoint to verify it is reachable.
6. The system saves the microservice details to the catalog database.
7. The system updates the dependency graph and begins the 30-second interval polling.
8. The system displays a Service Successfully Registered confirmation message.

Alternate Flow (5a. Endpoint Unreachable):

- 5a1. The system's initial ping to the provided health endpoint fails or times out.
- 5a2. The system halts the registration process and displays an error message: "Endpoint Unreachable. Please verify the URL or ensure the service is running."
- 5a3. The user is prompted to correct the URL or cancel the registration.